Debjeet Mazumder

Kolkata, West Bengal
KingDev4522 Debjeet-Mazumder
debjeetmazumdar3232@gmail.com +91 90078 97751

Professional Summary

B.Tech CSE student,3x hackathon laureate, and full-stack developer specializing in scalable, end-to-end systems. Experienced in architecting low-latency IoT ecosystems, multi-agent AI pipelines, and AI-driven B2B SaaS platforms. Proficient across embedded C++, robust Python backends, and high-performance interfaces to solve complex real-world engineering challenges.

Skills & Achievements

- **Languages:** Python, Java, C/C++, HTML5, CSS3
- **Applied Experience & Tools:** JavaScript, Kotlin, Dart, React.js, Next.js, Three.js, Git, GitHub
- **Frameworks & Libraries:** Flutter, Flask, GSAP, Jinja2
- **Core Concepts & Databases:** OOP, Data Structures & Algorithms, MySQL, SQLite, SQLAlchemy
- **Design & Modeling:** AutoCAD, Blender, Adobe Photoshop, Spline
- **Achievements:** JISTech 2K26: 1st place - App-Dev Competition, 2nd Runner-Up - Theme Based Project (AES), 3rd Runner-Up - Crazy Build 2K26, **Winner (Best Beginner Track)** - HexaFalls 2 Hackathon

Top Projects

Maargdarshan | AI-Enabled Smart Navigation Ecosystem — *C/C++, Flutter, Kotlin, Python, JavaScript, Dart, ESP32, Arduino*
[Team Project - Founder, Solo Architect & Lead Developer]
Architecture: Three-Tier Ecosystem | Website | App *(*Prototypes in active Beta testing)*

- **Hardware Node:** Self built and engineered a custom wearable IoT device integrating a multidirectional spatial awareness array for real-time environmental mapping and localized haptic response.
- **Web Backend:** Architected a low-latency web portal featuring a digital twin interface to visually track hardware orientation, integrated with real-time optical telemetry, historical spatial-analysis logs alongside emergency dispatch protocol.
- **Frontend & UI Engineering:** Engineered a high-performance, visually thematic UI utilizing advanced rendering techniques to achieve fluid physics (satin cloth and water ripple effects). Integrated dynamic thematic motifs (peacock feather cursor) to visually translate the project's assistive philosophy into an interactive digital experience.
- **Mobile Application:** Developed a memory-optimized native app in a headless ambient state, seamlessly unifying voice-navigated spatial AI, optical telemetry, and spatial-analysis logs with an OS-bypassing emergency routing protocol.
- **Backend Routing Server:** Architected the central backend API to bridge the entire ecosystem, handling real-time data flow, sensor telemetry routing, SOS dispatch routing,live optical telemetry streaming with low latency and state synchronization between the IoT hardware, mobile application, and web portal.

Patronus | Multimodal AI Forensics Ecosystem — *PyTorch, OpenCV, SQLite, ASGI, Zustand*
[Team Project - Co-Founder, Backend & Frontend Co-Developer, AI Co-Engineer]

- **5-Module AI Architecture & Tensor Physics:** Engineered specialized deep-learning evaluation pipelines across all 5 forensic modules (deploying YOLOv8, ResNet, and PyTorch Mel-spectrograms). Simultaneously developed the Luminescence engine, utilizing `adaptive_max_pool2d` tensors and Min-Max Normalization to mathematically preserve and amplify microscopic deepfake edge-signals within compressed media.
- **Full-Stack System Integration:** Connected the entire AI ecosystem to the frontend DOM by structuring strict SQLite database write-schemas across all 5 pipelines. Mapped complex Zustand JSON payloads to enable the immediate 60FPS rendering of live Base64 heatmaps and model verdicts.

City Multi-Specialty Clinic | CLI System — *C, File I/O, Data Structures*
[Team Project - Founder, Lead Architect & Lead Backend Engineer]

- Architected the core algorithmic workflows and memory-safe relational flat-file database engines (CRUD) in C, featuring custom Hex-XOR encryption for secure multi-role authentication.
- Engineered the dynamic clinical routing system, designing complex logic pipelines to seamlessly integrate patient triage, lab diagnostics, automated billing, and role-based data synchronization.

Chanakya AI | Real-Time Competitive Intelligence Engine — *Python, FastAPI, Crawl4AI, React.js, Tailwind CSS, Next.js*
[Team Project - Founder & Leader, Core Backend Developer & Logic Architect, Lead Designer]

- Engineered an enterprise-grade AI competitive intelligence platform, designing the overarching UI/UX blueprints and the immersive Hacker Theatre dashboard. Also designed an animated 3D cube on spline and integrated in the landing page.
- Architected the core backend logic, programming the exact data flow, context accumulation, and execution sequences across a 5-agent Gemini 3.5 Flash pipeline.
- Built the central data processing algorithms that autonomously evaluate disruptive business frameworks and map self-company metrics in real-time.

Smart Bluetooth 4WD RC Car — *C++, Arduino, L293D, HC-05, Bluetooth SPP*
[Solo Project - Hardware Builder & Firmware Developer]

- Engineered a rigid 6mm wooden chassis (20x12 cm) with a repurposed toy car canopy. Integrated an Arduino Uno, HC-05, and L293D shield, resolving motor jamming via iterative stress testing.
- Programmed embedded C++ differential drive (skid-steering) for real-time smartphone control, implementing custom PWM speed reduction to mitigate total system voltage brownouts under heavy load.

Education

B.Tech in Computer Science and Engineering (2025-2029)
JIS College of Engineering, Kalyani — *Pursuing*

Interests

- Participating in hackathons and tech events
- Full-stack web/mobile development, SaaS projects, Hardware & IoT integration